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Unconventional New WHP Design Using Advanced SRA and Data Driven Techniques Enhancing Knowledge from Green Field Design, Inspection thru Decommissioning (STR 5.0 Plus)

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Abstract

This paper introduces innovative approaches and methodologies in Wellhead Platform (WHP) structure design, enhancing the economic viability of marginal field developments. It presents breakthrough structural design advancements for offshore WHPs in water depths of 50-80 meters, utilizing novel approaches and data-driven techniques. Advanced Structural Reliability Analysis (SRA) was adopted as the state-of-the-art approach to ensure that overall structure integrity is maintained as As Low As Reasonably Practicable (ALARP) under fit for service concept while certain design parameters were optimized to meet the target of weight reduction. As stated by ISO19901-9, SRA is the most complex method but provides the most reducing conservatism and it is categorized as level 5.0 for quantitative ultimate strength assessment approach. To make it as level 5.0 plus, the historical inspection data from over four decades and lessons learned from recent decommissioned assets were deeply investigated to derive new design parameters. The project demonstrates significant life cycle cost reduction, unconventionally achieving significant weight minimization and at least 30% saving in resources in comparison to the latest project in Gulf of Thailand (GoT) region. These unconventional approaches contribute to the broader petroleum industry by setting new standards for cost-effective and sustainable offshore platform design. By implementing the new STR 5.0 plus concept, the total platform life cycle cost could be reduced by 10% compared to previous platform generations. Total weight reduction is 278 MT, including 89 MT from the topside (12% reduction) and 189 MT from the substructure (15% reduction). A Gas WHP topside weight of 636 MT and substructure weight of 1,076 MT makes this new design the leanest WHP of its kind for 70m water depth in the GoT region. With Not-Normally-Manned Installations (NNMI) requirement under L-2 exposure level, key achievements include: 1) Deck area reduced over 10% totally; 2) Jacket bays reduced to 4 bays; 3) Adopting 25-year return period extreme storm conditions; 4) Best estimate soil saves pile materials; 5) SRA in the green field stage provides not only predictable probability of failures of the platform during its design life but also provides an efficient integrity management program without OPEX impact while CAPEX is reduced; and 6) Nearly 30% sacrificial anodes reduced. In sum, the platform probability of failure from all potential degradation mechanisms was rigorously assessed and ensured to be well-designed and controlled to be

within the allowable risk threshold (1×10^{-3}) throughout the designated design life with adequate and proper recommended inspection program.

Introduction

The increasing demand for energy and the depletion of small-to-medium, easily accessible hydrocarbon reservoirs in GoT region have driven the petroleum industry towards the development of marginal fields. These marginal fields, often characterized by smaller reserves and challenging environmental conditions, necessitate innovative and cost-effective solutions for their infrastructure. Traditional offshore Wellhead Platform (WHP) designs, while robust, can be prohibitively expensive for such developments, impacting their economic viability. This paper introduces a promising approach in WHP design, moving beyond conventional prescriptive methods to embrace advanced **Structural Reliability Analysis (SRA)** and **data-driven techniques**. The objective is to achieve significant life cycle cost reductions, weight minimization, and enhanced sustainability, thereby unlocking the potential of marginal fields in regions like the GoT.

The conventional approach to offshore platform design often relies on deterministic methods and conservative safety factors, which can lead to over-designed structures. While ensuring safety, the conventional approach can result in excessive material usage, higher fabrication costs, and prolonged project schedules. In contrast, SRA provides a probabilistic framework for assessing structural integrity throughout its design life, allowing for a more refined understanding of risks and uncertainties. By quantifying the probability of failure, engineers can optimize designs to meet specific reliability targets, leading to more efficient and economical structures without compromising safety.

Table 1—Quantitative Ultimate Assessment Method Selection by ISO 19901-9

Level	Design Approach	Description
Level 1	Linear-elastic analysis	Basic analysis assuming linear elastic material behavior.
Level 2	Linear-elastic redundancy	Design incorporating redundancy based on linear elastic principles.
Level 3	Static non-linear analysis	Analysis considering non-linear material behavior under static loads.
Level 4	Dynamic non-linear analysis	Advanced analysis considering non-linear material behavior under dynamic loads.
Level 5	Structural reliability analysis	Comprehensive analysis to quantify the probability of structural failure, incorporating uncertainties.

Furthermore, the integration of data-driven techniques is crucial for this unconventional design philosophy. Decades of operational data, including inspection records and lessons learned from decommissioned assets, offer invaluable insights into the actual performance and degradation mechanisms of offshore structures. By leveraging this "data lake," engineers can move beyond theoretical assumptions and incorporate real-world experience into the design process in which never been conducted before in GoT. This paper details how this rich historical data, combined with advanced SRA, has led to the development of the STR 5.0 Plus concept, a breakthrough in WHP design.

The STR 5.0 Plus concept aims to address the challenges of marginal field development by focusing on lean design principles and optimized integrity management. This includes the adoption of a Not-Normally Manned Installation (NNMI) concept under L-2 exposure level, a reduced number of jacket bays for

below 70m water depth, and the application of a 25-year return period for extreme storm conditions, all underpinned by rigorous probabilistic analysis. The goal is to establish new benchmarks for cost-effectiveness and sustainability in offshore platform design, setting a precedent for future developments in the petroleum industry.

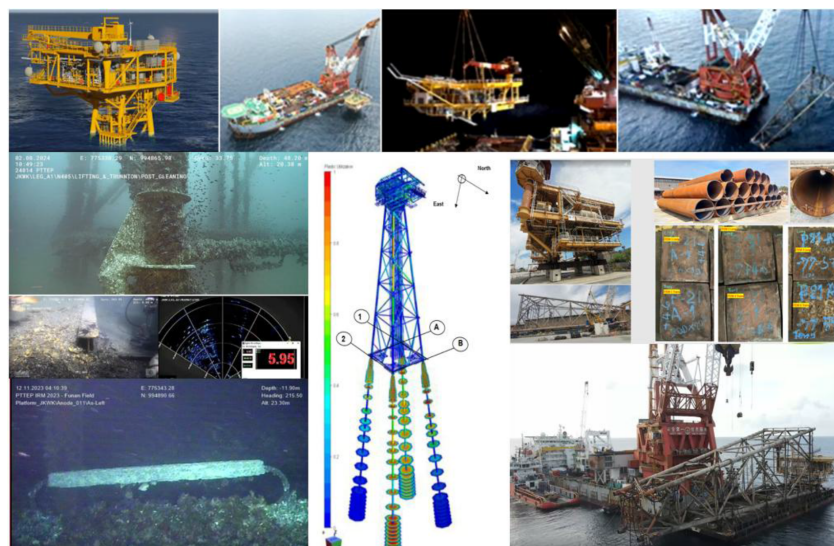


Figure 1—Life cycle of Oil and Gas Wellhead Platform: Engineering design, Installation, Inspection & Decommission

Problem Statement

The traditional approach to designing Wellhead Platforms (WHPs) for offshore oil and gas fields, particularly in regions like the Gulf of Thailand (GoT) with scattered small prospects and unique reservoir characteristics, often leads to designs that are not optimally aligned with the economic realities of marginal field development. Current practices, heavily reliant on prescriptive codes, generalized company guidelines and deterministic safety factors, tend to result in over-designed structures. This conservatism, while ensuring safety, translates directly into higher material consumption, increased fabrication complexity, extended project schedules, and ultimately, a higher total installed cost.

Specifically, several key challenges necessitate a re-evaluation of conventional WHP design:

- Economic Viability of Marginal Fields:** The inherent smaller reserves and often challenging operational environments of marginal fields make them highly sensitive to capital expenditure (CAPEX) and operational expenditure (OPEX). Conventional WHP designs, with their substantial weight, facilities and material requirements, can render these fields economically unviable. There is a pressing need for designs that significantly reduce CAPEX without compromising safety or long-term integrity.
- Suboptimal Integrity Management:** While existing integrity management programs are in place, they often follow fixed, time-based inspection schedules or semi-quantitative approach was adopted by following risk ranking mechanism. These schedules may not fully account for the actual degradation behavior of structures or leverage the wealth of historical inspection data. This can lead to either unnecessary inspections (increasing OPEX) or, conversely, missed opportunities for early intervention in critical areas.
- Environmental and Sustainability Concerns:** The carbon footprint associated with material production, fabrication, and transportation of heavy offshore structures is significant. A lean design that minimizes weight and resource consumption contributes directly to the industry's sustainability goals.

- **Inefficient Resource Utilization:** Over-design leads to an inefficient use of steel and other resources. This not only impacts project costs but also has environmental implications, contradicting the industry's growing emphasis on sustainable practices.
- **Lack of Data-Driven Feedback Loop:** Despite decades of operational experience and extensive inspection records, there is often a disconnect between the data gathered during a platform's life cycle and its direct application in the initial design phase of new structures. Lessons learned from decommissioned assets, particularly regarding actual material degradation, fatigue performance, and anode depletion, are not systematically fed back into the design parameters for new "greenfield" projects.
- **Adaptation to Evolving Metocean Data:** Metocean conditions can change over time, and relying solely on historical design parameters without incorporating the latest data can lead to either overly conservative or potentially under-designed elements. A dynamic approach that integrates updated metocean information is essential.

The core problem, therefore, is the need to bridge the gap between traditional, conservative design methodologies and the imperative for more economically viable, resource-efficient, and intelligently managed WHP structures for marginal field developments. This requires a shift towards a design philosophy that is inherently probabilistic, data-informed, and focused on optimizing the entire life cycle of the asset, from design through decommissioning. The STR 5.0 Plus concept directly addresses these challenges by integrating advanced SRA and comprehensive data analysis to deliver a truly unconventional and optimized WHP design.

Methodology

To achieve the optimization target of condensed layout and weight reduction with single standardized design during Engineering phase, project team adhered to the Scope of Requirements (SOR) obtained from asset planning team. For structural engineering, the design data and analysis approaches have been revisited thoroughly by several out-of-the-box means to overcome the project challenges. Introducing design thinking workshops with multidiscipline SMEs and brainstorming with external engineering firm were adopted to list out the potential initiatives. Project team has conducted the reviewing of the applicable international codes, Company Engineering Guideline Specification (PEGS) and previous best practices in WHP design across GoT assets, which possess difference from corporate traditional ways.

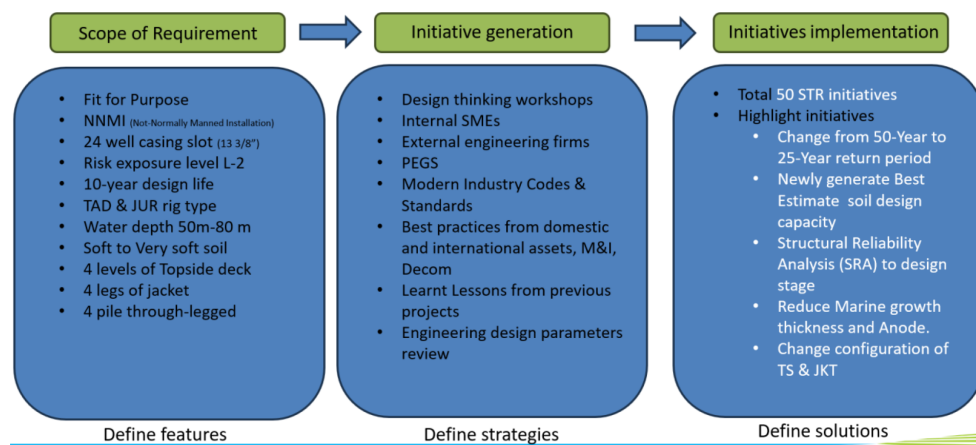


Figure 2—Proposed initiatives adoption

Data driven techniques enhancing knowledge from green field design, Inspection thru Decommissioning

To promote the Data-Driven technique incorporating the new green field practice, inspection and decommissioning best practices into new WHP engineering design phase for structure sizing optimization, all potential initiatives and existing data reservoirs in the company have been considered and addressed in the structural analysis. Then, the SRA was the tool to ensure that the overall integrity of WHP can be managed throughout the design life after the selected initiatives were adopted accordingly.

Highlight structural design changes from previous project are itemized as follows.

Metoccean data & Design return period.

Table 2—Design return period

Previous project	STR 5.0+ project
Operating Storm, return period of 1-year shall be used for the platform design (Jacket, Topside & Pile) under normal operating, partial standby drilling, well servicing and soft mooring conditions.	Operating Storm, return period of 1-year shall be used for the platform design (Jacket, Topside & Pile) under normal operating, partial standby drilling, well servicing and soft mooring conditions.
Extreme Storm, return period of 50-year shall be used for platform design (Jacket, Topside & Pile) with full stand-by drilling rig conditions.	Extreme Storm, return period of 25-year shall be used for platform design (Jacket, Topside & Pile) with full stand-by drilling rig conditions.
-	Extreme Storm, return period of 50-year shall be used for the platform design Jacket, Topside & Pile without drilling rig condition.
Extreme Storm, return period of 50-year shall be used for the topside design under well servicing condition.	Extreme Storm, return period of 50-year shall be used for the topside design under well servicing condition.
Pile penetration depths shall be determined based on 1-year & 50-year storm conditions mentioned above and 100-year return period extreme storm.	Pile penetration depths shall be determined based on 1-year, 25-year & 50-year storm conditions mentioned above and 100-year return period extreme storm.
Transportation analysis for topside and jacket shall be based on 10-year return period.	Transportation analysis for topside and jacket shall be based on 10-year return period.

Best estimate soil. The best estimate soil was newly developed for substructure design. It is obtained by finding PY curve that is close to the median value from various available PY data. The 1st screening is to verify PY data up 100m depth to assess general pile penetration depth and 2nd screening is to verify PY data up to 35m where the pile bending moment normally occurred.

Based on the screening results of PY data between mudline to 100m depth, it shows borehole of selected WHP in Area B2 is the nearest to the median value. From the screening results of PY data from mudline to 35m, the nearest PY data to median value is also borehole of selected WHP in Area B2.

Thus, it can be concluded that soil borehole of selected WHP in Area B2 can be considered as the "best estimate soil model" for substructure generic design. Once the site-specific soil data is obtained, it shall be used to cross check the member sizing adequacy.

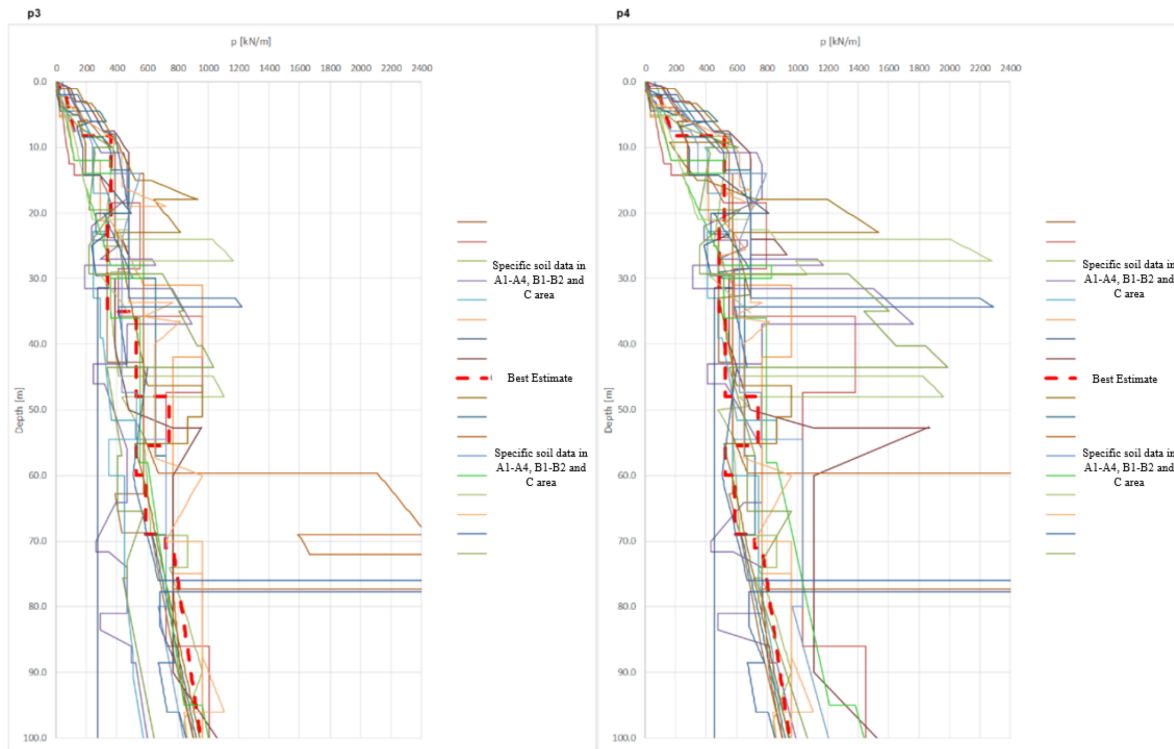


Figure 3—Best estimate soil generated from updated soil data sets

Configuration of topside and jacket. The intercoastal beam configuration was chosen for topside. The jacket was reduced from 5 bays to 4 bays for 70m water depth case.



Figure 4—Jacket configuration change (red 5 bays, yellow 4 bays)

Marine Growth (MG) and Anode quantity. The marine growth thickness profile was newly redefined by calibrating the generalized company guideline with the actual measurement value obtained from ROV special tools which its data lake was documented since 2009 up to 2024. The cathodic protection system using sacrificial anode was also rigorously recalculated by considering the recent actual weighing of decommissioned anodes in 2023 and subsea inspection database recorded since 2007 up to 2024.

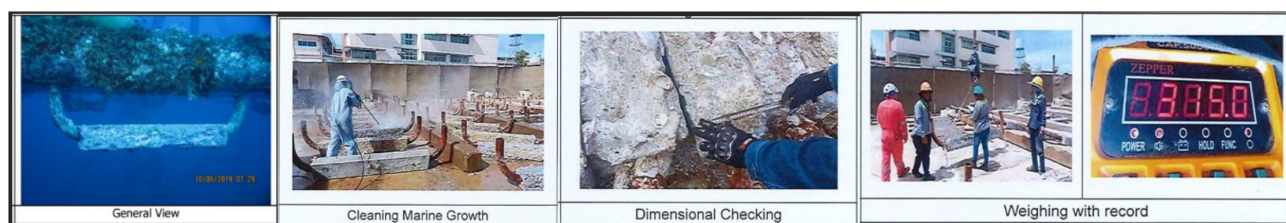


Figure 5—Anode subsea inspection (Left), Anode weighing (Right)

Structural Reliability Analysis

Structure Integrity Management (SIM) system with Risk-Based Inspection (RBI) approach, in accordance with API RP 2SIM, is generally adopted to maintain the integrity of existing offshore structure throughout its design service life. Project team has newly introduced SRA for new platform design purpose in compliance with API RP 2A, ISO 19902 and DNV RP C210. The acceptance criteria of allowable threshold have followed the relevant company guidelines and industry codes.

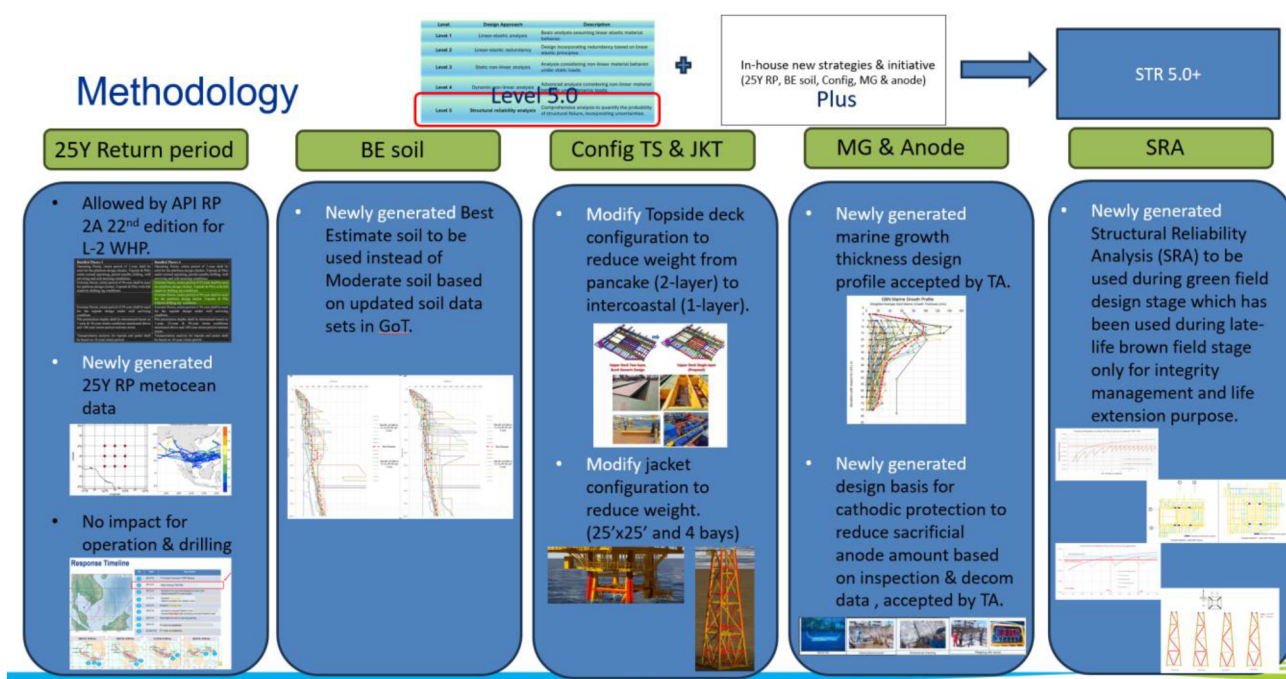


Figure 6—Highlight initiatives for WHP with SRA adopted in STR5.0+ concept

The following effects of the system probability of failure are considered in SRA study:

- Member overstressing due to environmental overloading with and without redundancy analysis
- Fatigue degradation
- Member corrosion (both in splash zone and below the splash zone)
- Increase in scour depth beyond the design Scour
- Increase in marine growth profile beyond the design profile
- Undiscovered member loss/indentation due to ship impact/dropped object
- Undiscovered member failure due to component overloading by the impact of corrosion

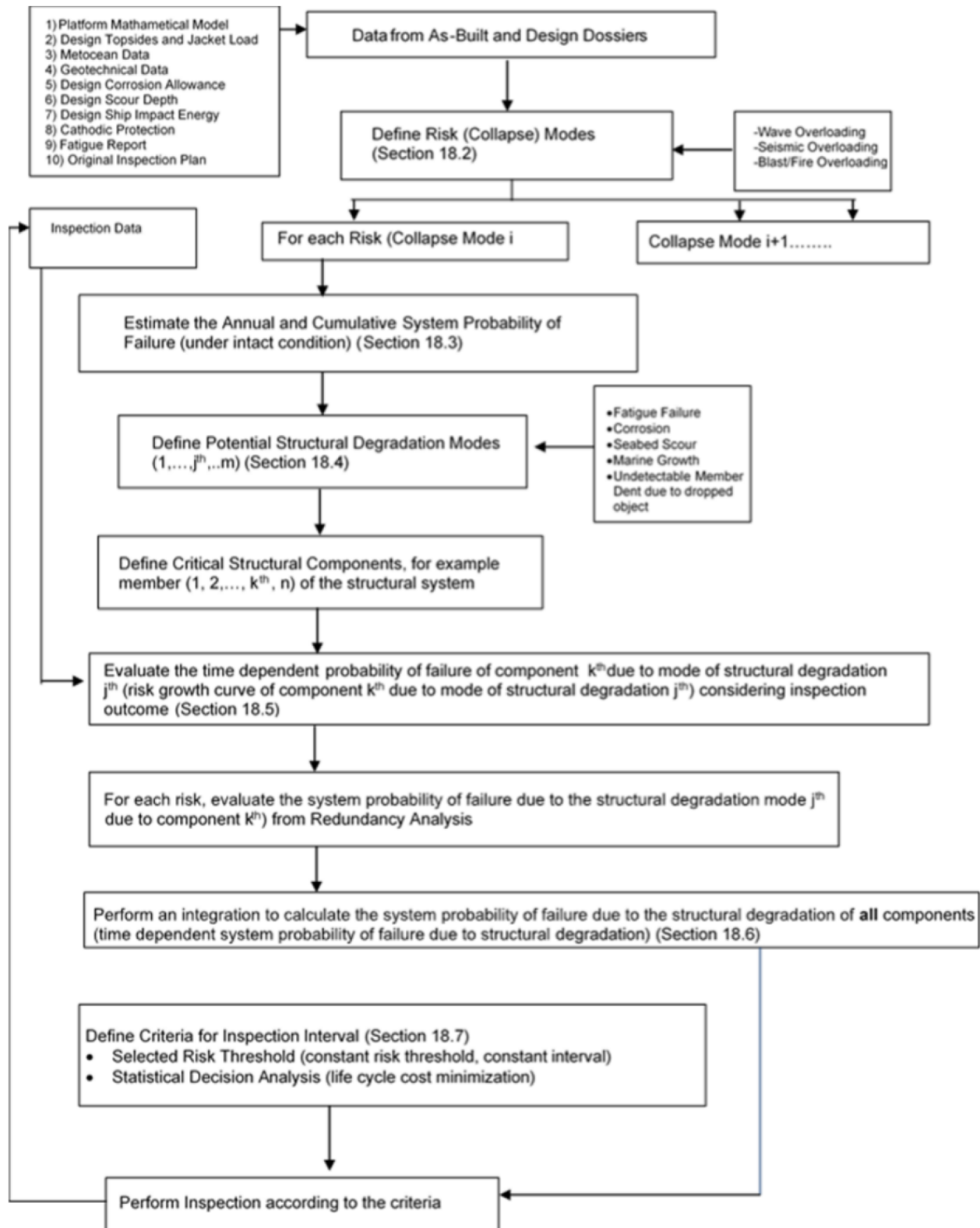


Figure 7—SRA work flow

Result

SRA results show the design jacket with generic best estimate soil meets the failure probability threshold throughout its service life. The first inspection should occur within 10 years of installation. If no major anomalies are found, the next inspection should be within 6 years. Level 3 inspections should use Flooded

Member Detection (FMD) and Closed Visual Inspection (CVI) on recommended structural members and joints.

Table 3—Inspection schedule program

Vintage Year	Inspection Campaign	Base Line Inspection	1 st Periodic Inspection	2 nd Periodic Inspection (if results from 1 st Periodic Inspection are favorable)
Year 0 (Year of Installation)	When	Within 12 Months after installation	10 year	6 Year after 1 st periodic inspection
	Activities Carried Out	Level 1 & 2	Level 1 & 2, Level 3 FMD on selected members	Level 1 & 2, Level 3 FMD on selected members

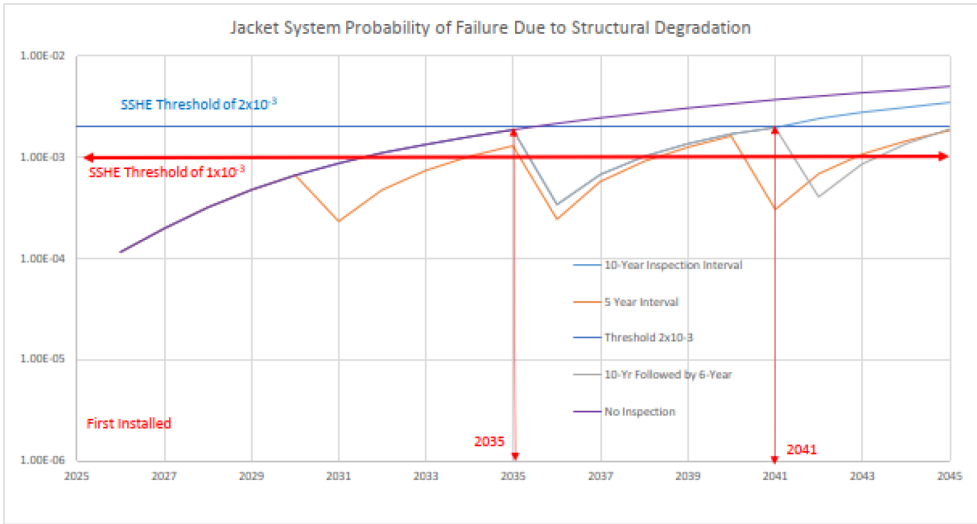


Figure 8—Jacket probability of failure against service year

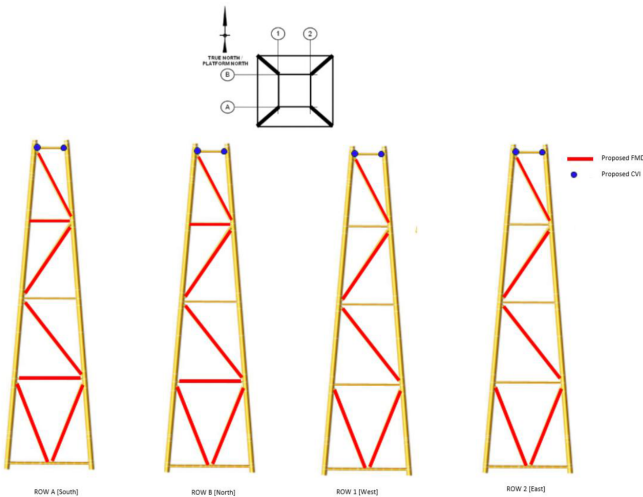


Figure 9—Jacket inspection location and NDT technique

Conclusion

This paper presents the STR 5.0 Plus concept, an unconventional approach to Wellhead Platform (WHP) design that leverages advanced Structural Reliability Analysis (SRA) and extensive data-driven techniques to enhance the economic viability and sustainability of marginal field developments. By moving beyond

traditional deterministic methods, this methodology provides a robust framework for optimizing WHP designs from inception through decommissioning.

The implementation of the STR 5.0 Plus concept has demonstrated significant benefits, including a 10% reduction in total platform life cycle cost, a significant weight minimization (saving 89 MT or 12% from topside, saving 189 MT or 15% from substructure), and a 30% saving in resources. Key design innovations, such as the adoption of a Not-Normally Manned Installation (NNMI) concept, a reduced number of jacket bays, and the application of a 25-year return period for extreme storm conditions, have contributed to making this new design the leanest WHP of its kind in the SEA region. The rigorous application of SRA in the green field stage not only predicts the probability of platform failures but also facilitates an efficient integrity management program that reduces CAPEX without impacting OPEX. Furthermore, the optimization efforts resulted in a nearly 30% reduction in sacrificial anodes.

A core outcome of this study is the development of a refined inspection planning strategy based on the probabilistic assessment of various degradation mechanisms. The analysis confirms that the platform's probability of failure from all potential degradation mechanisms remains well within the allowable risk threshold (1×10^{-3}) throughout its designated design life, provided the recommended inspection programs are adequately implemented.

Based on the platform reliability updating analysis, the following inspection planning is proposed:

- **Baseline Inspection:** To be completed within 12 months after platform installation.
- **Periodic Underwater Inspection Campaign (Jacket Structures):**
 - The first inspection is recommended approximately **10 years after installation**.
 - Subsequent inspections are proposed at **6-year intervals** after the first periodic inspection, assuming favorable outcomes (no major anomalies detected).
 - These inspections will include Level 1 & 2 activities, and Level 3 Flooded Member Detection (FMD) on selected members
- **Periodic Above-water Inspection Campaign (Topsides Structures):**
 - Inspections should be conducted at **2-year intervals** until the service life reaches 10 years.
 - Thereafter, **annual inspections** are recommended until the end of the service life.
 - These activities will include Level 1 & 2 inspections, and Close Visual Inspection (CVI) at proposed member/joint locations. Non-destructive testing (NDT) is only required if CVI identifies anomalies.

The quantitative risk-based inspection performed in this study primarily emphasizes degradation due to fatigue deterioration, undiscovered dropped objects, and ship impacts on system robustness. Other risks, such as those from corrosion, scour, and marine growth, are considered of secondary significance due to effective design allowances and historical inspection data.

This unconventional design philosophy, underpinned by advanced SRA and data-driven insights, sets a new standard for cost-effective, sustainable, and reliable offshore platform development, particularly for marginal fields. It demonstrates that a proactive, risk-informed approach to design and integrity management can yield substantial economic and operational advantages.

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